**Kalman Filter Sensor Fusion**
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The Kalman filter (*Kalman*, 1960) and its non-linear extensions—the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF)—provide the theoretical and computational framework for optimal recursive fusion of these diverse sensor streams. The result is a posterior distribution over the vehicle state that is both more accurate and more robust than any individual measurement.

When all sensor models are linear, the standard Kalman filter applies. For fusing $S$ sensors with observation models:

$$\mathbf{z}_t^{(s)} = \mathbf{H}^{(s)}\mathbf{x}_t + \mathbf{v}_t^{(s)}, \qquad \mathbf{v}_t^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}^{(s)})$$

Under the conditional independence assumption, the joint update is equivalent to performing $S$ sequential scalar (or block) updates, one sensor at a time. This is computationally advantageous when each sensor has a small-dimensional measurement. Predict:

$$\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}$$
$$\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^\top + \mathbf{Q}$$

Update for sensor $s$ (iterate $s = 1, \ldots, S$):

$$\boldsymbol{\nu}_t^{(s)} = \mathbf{z}_t^{(s)} - \mathbf{H}^{(s)}\hat{\mathbf{x}}_{t|t-1}$$
$$\mathbf{S}_t^{(s)} = \mathbf{H}^{(s)}\mathbf{P}_{t|t-1}(\mathbf{H}^{(s)})^\top + \mathbf{R}^{(s)}$$
$$\mathbf{K}_t^{(s)} = \mathbf{P}_{t|t-1}(\mathbf{H}^{(s)})^\top(\mathbf{S}_t^{(s)})^{-1}$$
$$\hat{\mathbf{x}}_{t|t} \leftarrow \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t^{(s)}\boldsymbol{\nu}_t^{(s)}$$
$$\mathbf{P}_{t|t} \leftarrow (\mathbf{I} - \mathbf{K}_t^{(s)}\mathbf{H}^{(s)})\mathbf{P}_{t|t-1}(\mathbf{I} - \mathbf{K}_t^{(s)}\mathbf{H}^{(s)})^\top + \mathbf{K}_t^{(s)}\mathbf{R}^{(s)}(\mathbf{K}_t^{(s)})^\top$$

At each sequential update, the current $\hat{\mathbf{x}}_{t|t}$ and $\mathbf{P}_{t|t}$ replace $\hat{\mathbf{x}}_{t|t-1}$ and $\mathbf{P}_{t|t-1}$ for the next sensor.

*Theorem (Sequential equivalence)*: Under the conditional independence assumption, the joint update from all $S$ sensors simultaneously is algebraically equivalent to $S$ sequential updates in any order. This follows from the block matrix inversion lemma applied to the stacked observation model:

$$\begin{pmatrix} \mathbf{z}_t^{(1)} \\ \vdots \\ \mathbf{z}_t^{(S)} \end{pmatrix} = \begin{pmatrix} \mathbf{H}^{(1)} \\ \vdots \\ \mathbf{H}^{(S)} \end{pmatrix} \mathbf{x}_t + \begin{pmatrix} \mathbf{v}_t^{(1)} \\ \vdots \\ \mathbf{v}_t^{(S)} \end{pmatrix}, \quad \mathbf{R} = \mathrm{diag}(\mathbf{R}^{(1)}, \ldots, \mathbf{R}^{(S)})$$

The block-diagonal structure of $\mathbf{R}$ is the key condition. If sensors were correlated, $\mathbf{R}$ would have off-diagonal blocks and joint update would be required.

<u>The Extended Kalman Filter (EKF)</u>

The CTRV motion model and the RADAR measurement model (which expresses range and bearing as non-linear functions of *Cartesian* state) are both non-linear. The Extended Kalman Filter handles

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non-linearity by linearizing the system equations around the current state estimate via a first-order Taylor expansion.

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t) + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$$
$$\mathbf{z}_t^{(s)} = h^{(s)}(\mathbf{x}_t) + \mathbf{v}_t^{(s)}, \qquad \mathbf{v}_t^{(s)} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}^{(s)})$$

Define the state transition *Jacobian*:

$$\mathbf{F}_t = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}_{t|t}} \in \mathbb{R}^{n\times n}$$

and the observation *Jacobian* for sensor s:

$$\mathbf{H}_t^{(s)} = \left.\frac{\partial h^{(s)}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}_{t|t-1}} \in \mathbb{R}^{m_s\times n}$$

A RADAR sensor located at the origin measures the range $r$, azimuth bearing $\phi$, and range rate $\dot{r}$ of a target at state $(x, y, \dot{x}, \dot{y})^\top$:

$$h_{\mathrm{RADAR}}(\mathbf{x}) = \begin{pmatrix} r \\ \phi \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \sqrt{x^2+y^2} \\ \arctan(y/x) \\ \dfrac{x\dot{x}+y\dot{y}}{\sqrt{x^2+y^2}} \end{pmatrix}$$

The corresponding observation *Jacobian* is:

$$\mathbf{H}_{\mathrm{RADAR}} = \begin{pmatrix} \frac{x}{r} & \frac{y}{r} & 0 & 0 \\ -\frac{y}{r^2} & \frac{x}{r^2} & 0 & 0 \\ \frac{y(\dot{x}y-\dot{y}x)}{r^3} & \frac{x(\dot{y}x-\dot{x}y)}{r^3} & \frac{x}{r} & \frac{y}{r} \end{pmatrix}$$

where $r = \sqrt{x^2+y^2}$.

Angle wrapping: The bearing $\phi = \arctan(y/x)$ lies in $(-\pi, \pi]$. The innovation $\boldsymbol{\nu}_\phi = \phi_{\mathrm{meas}} - \phi_{\mathrm{pred}}$ must be normalized to $(-\pi, \pi]$ to avoid jumps.

EKF Predict:

$$\hat{\mathbf{x}}_{t|t-1} = f(\hat{\mathbf{x}}_{t-1|t-1}, \mathbf{u}_{t-1})$$
$$\mathbf{P}_{t|t-1} = \mathbf{F}_{t-1}\,\mathbf{P}_{t-1|t-1}\,\mathbf{F}_{t-1}^\top + \mathbf{Q}$$

EKF Update (sensor s):

$$\boldsymbol{\nu}_t^{(s)} = \mathbf{z}_t^{(s)} - h^{(s)}(\hat{\mathbf{x}}_{t|t-1})$$
$$\mathbf{S}_t^{(s)} = \mathbf{H}_t^{(s)}\,\mathbf{P}_{t|t-1}\,(\mathbf{H}_t^{(s)})^\top + \mathbf{R}^{(s)}$$

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$$\mathbf{K}_t^{(s)} = \mathbf{P}_{t|t-1}\,(\mathbf{H}_t^{(s)})^\top\,(\mathbf{S}_t^{(s)})^{-1}$$
$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t^{(s)}\,\boldsymbol{\nu}_t^{(s)}$$
$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t^{(s)}\,\mathbf{H}_t^{(s)})\,\mathbf{P}_{t|t-1}$$

Limitation: The EKF is a first-order approximation. When f or h are highly non-linear, linearization may be inaccurate, leading to filter inconsistency (the true state lies outside the $3\sigma$ bounds more than 0.3% of the time) or divergence. The UKF addresses this.

<u>Unscented Kalman Filter (UKF)</u>

The Unscented Transform (Julier & Uhlmann, 1997) is a method for computing the mean and covariance of a random variable that has been passed through a non-linear function, without computing Jacobians. It relies on a deterministic set of sigma points $\{\boldsymbol{\mathcal{X}}_i\}$ that capture the first two moments (mean and covariance) of the prior distribution exactly.

Given a Gaussian random variable $\mathbf{x} \sim \mathcal{N}(\bar{\mathbf{x}}, \mathbf{P})$ with $\mathbf{x} \in \mathbb{R}^n$, choose $2n+1$ sigma points:
$$\boldsymbol{\mathcal{X}}_0 = \bar{\mathbf{x}}$$
$$\boldsymbol{\mathcal{X}}_i = \bar{\mathbf{x}} + \left(\sqrt{(n+\lambda)\mathbf{P}}\right)_i, \quad i = 1, \ldots, n$$
$$\boldsymbol{\mathcal{X}}_{n+i} = \bar{\mathbf{x}} - \left(\sqrt{(n+\lambda)\mathbf{P}}\right)_i, \quad i = 1, \ldots, n$$
where $\left(\sqrt{(n+\lambda)\mathbf{P}}\right)_i$ denotes the $i$–th column of the matrix square root (*Cholesky* factor), and $\lambda = \alpha^2(n+\kappa) - n$ is a composite scaling parameter.

UKF tuning parameters:
- $\alpha \in (0, 1]$: controls the spread of sigma points; typically $\alpha = 10^{-3}$.
- $\kappa \geq 0$: secondary scaling; often $\kappa = 0$ or $3 - n$.
- $\beta \geq 0$: incorporates prior knowledge about the distribution; $\beta = 2$ is optimal for Gaussian.

Weights for mean and covariance reconstruction:
$$W_0^{(m)} = \frac{\lambda}{n+\lambda}, \qquad W_0^{(c)} = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta)$$
$$W_i^{(m)} = W_i^{(c)} = \frac{1}{2(n+\lambda)}, \quad i = 1, \ldots, 2n$$

*UKF Predict Step*
1. Compute sigma points from $\hat{\mathbf{x}}_{t-1|t-1}$ and $\mathbf{P}_{t-1|t-1}$.
2. Propagate each sigma point through $f$:
   $\boldsymbol{\mathcal{X}}_i^{=} f(\boldsymbol{\mathcal{X}}_i), \quad i = 0, \ldots, 2n$
3. Recover predicted mean and covariance:
   $$\hat{\mathbf{x}}_{t|t-1} = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\mathcal{X}}_i$$
   $$\mathbf{P}_{t|t-1} = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\mathcal{X}}_i^{-} \hat{\mathbf{x}}_{t|t-1})(\boldsymbol{\mathcal{X}}_i^{-} \hat{\mathbf{x}}_{t|t-1})^\top + \mathbf{Q}$$

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*UKF Update Step*

1. Re-compute sigma points from $\hat{\mathbf{x}}_{t|t-1}$ and $\mathbf{P}_{t|t-1}$.
2. Propagate through the observation function $h$:
   $\boldsymbol{\mathcal{Z}}_i = h(\boldsymbol{\mathcal{X}}_i), \quad i = 0, \ldots, 2n$
3. Compute predicted measurement mean and covariances:
   $$\hat{\mathbf{z}}_t = \sum_{i=0}^{2n} W_i^{(m)} \boldsymbol{\mathcal{Z}}_i$$
   $$\mathbf{S}_t = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\mathcal{Z}}_i - \hat{\mathbf{z}}_t)(\boldsymbol{\mathcal{Z}}_i - \hat{\mathbf{z}}_t)^\top + \mathbf{R}$$
   $$\mathbf{T}_t = \sum_{i=0}^{2n} W_i^{(c)} (\boldsymbol{\mathcal{X}}_i - \hat{\mathbf{x}}_{t|t-1})(\boldsymbol{\mathcal{Z}}_i - \hat{\mathbf{z}}_t)^\top$$
4. Compute Kalman gain and update:
   $\mathbf{K}_t = \mathbf{T}_t \mathbf{S}_t^{-1}$
   $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(\mathbf{z}_t - \hat{\mathbf{z}}_t)$
   $\mathbf{P}_{t|t} = \mathbf{P}_{t|t-1} - \mathbf{K}_t \mathbf{S}_t \mathbf{K}_t^\top$

The UKF achieves third-order accuracy for Gaussian inputs (vs. first-order for the EKF) and requires no *Jacobian* computation, making it preferable when the transition or observation functions are difficult to differentiate analytically.

<u>Concluding Remarks</u>

The table below provides a unified view of the three filter variants discussed:

| Property | Linear KF | EKF | UKF |
|---|---|---|---|
| Motion model | Linear $\mathbf{F}$ | Non-linear $h$, Jacobian $\mathbf{H}_t$ | Non-linear $f$, no Jacobian |
| Observation model | Linear $\mathbf{H}$ | Non-linear $h$, Jacobian $\mathbf{H}_t$ | Non-linear $h$, no Jacobian |
| Posterior approx. | Exact Gaussian | 1st-order Gaussian | 3rd-order Gaussian |
| Complexity per step | $O(n^3)$ | $O(n^3)$ | $O(n^3)$ |
| Jacobian required | No | Yes | No |
| Divergence risk | Low | Moderate (high non-linearity) | Lower |
| Typical use case | GPS+IMU (linear) | RADAR+GPS | CTRV with RADAR |

*The complete EKF cycle for autonomous driving sensor fusion*:

$$\hat{\mathbf{x}}_{t|t-1} = f(\hat{\mathbf{x}}_{t-1|t-1}), \quad \mathbf{P}_{t|t-1} = \mathbf{F}_t \mathbf{P}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}$$
$$\boldsymbol{\nu}_t = \mathbf{z}_t - h(\hat{\mathbf{x}}_{t|t-1}), \quad \mathbf{S}_t = \mathbf{H}_t \mathbf{P}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}$$
$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}$$
$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t, \quad \mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t)\mathbf{P}_{t|t-1}$$

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*Practical Considerations for Autonomous Driving Systems*

Fault detection and isolation: Sensors can fail (GPS spoofing, LiDAR obscuration by rain, RADAR false echoes). A gating function rejects measurements whose NIS exceeds a threshold:

$$\mathrm{NIS}_t > \chi^2_{m,1-\alpha} \implies \text{reject measurement}$$

Coordinate frames: Careful management of coordinate transformations is essential. All sensors must ultimately report in a common reference frame (e.g., a local East-North-Up frame centered at the vehicle's initial position). Errors in extrinsic sensor calibration — the rigid-body transformation between the sensor and the vehicle body frame — are a major practical source of systematic bias.

Computational constraints: Autonomous driving systems operate under hard real-time constraints. The EKF and UKF are both O(n^3) but with different constants. For high-dimensional states (e.g., n = 15 for full INS integration), the UKF requires 2n + 1 = 31 sigma point propagations versus a single linearization for the EKF. Square-root implementations of both filters improve numerical stability and run approximately twice as fast.

Beyond Kalman filtering: Modern autonomous driving stacks increasingly use factor graph optimization (e.g., GTSAM, g2o) and pose graph SLAM for mapping and localization, which jointly optimize a sliding window of past states. Kalman filters remain the standard for real-time state estimation of the ego-vehicle's motion.

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